

MEIOSIS

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The term 'meiosis' was given by Moore and Farmer (1905), though it was 1st of all ^{experimentally} demonstrated by Benden (1883) but its detail was described by Winiwates (1900) - (X) first described by Hertwig

Defn: meiosis is a type of karyokinesis (cell division) in which a very 1st chromosomes duplicate and then ~~reduced to half~~ just reduced to half to form 4-haploid non-identical daughter cells.

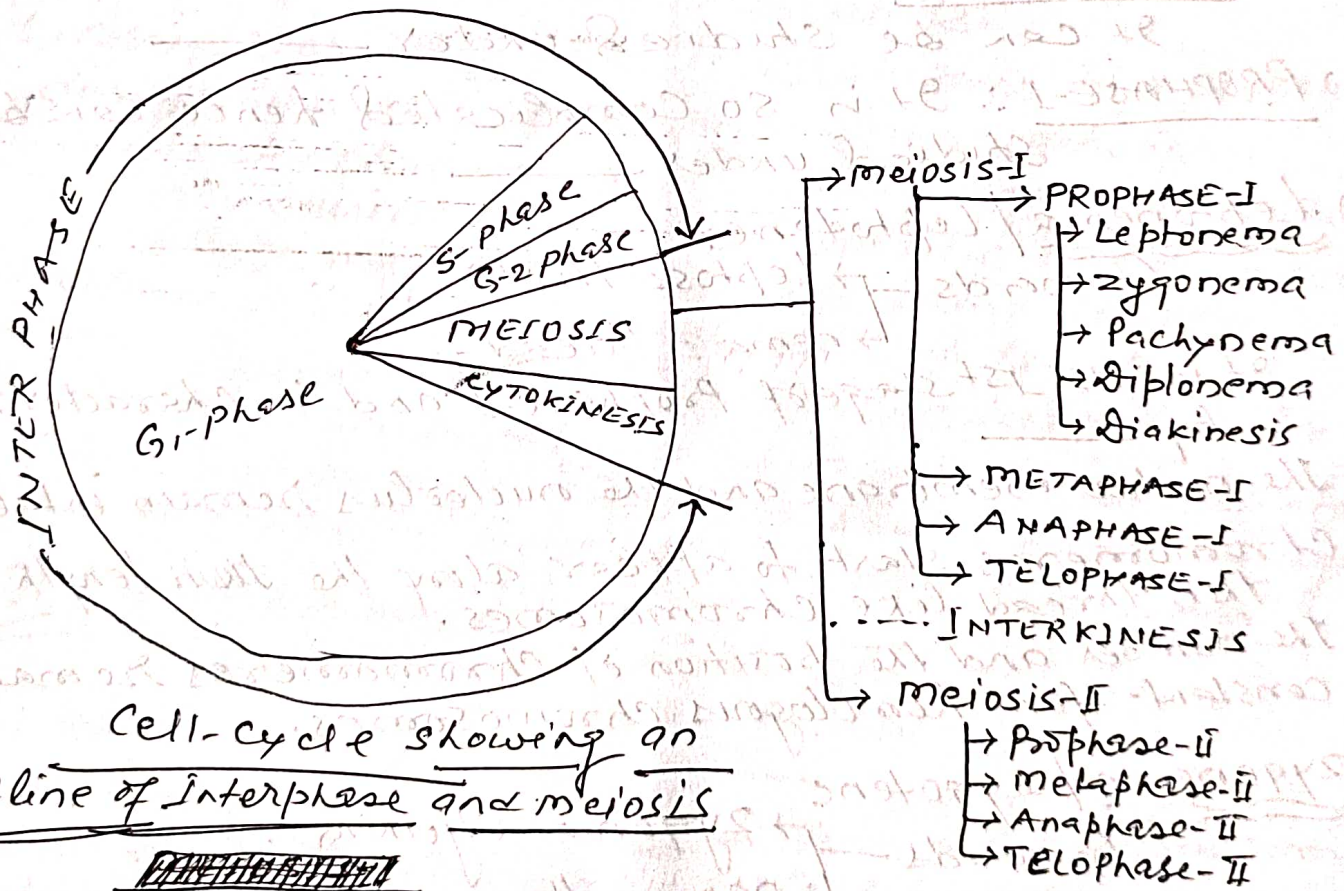
occurrence :- sexual cells, the cells in which meiosis occurs are referred as 'meiocytes' eg. in plant microspores, megaspores, etc

Actually in meiosis the nucleus and the cytoplasm divide twice while the chromosomes split longitudinally only once as a result four haploid non-identical daughter cells are formed. Meiosis involves two divisions —

(i) meiosis-I = Reductional division = Heterotypic division

(ii) meiosis-II = Equational division = Homotypic division

A continuously dividing cell passes through a period of gap known as INTERPHASE which is well illustrated in the following cell-cycle —



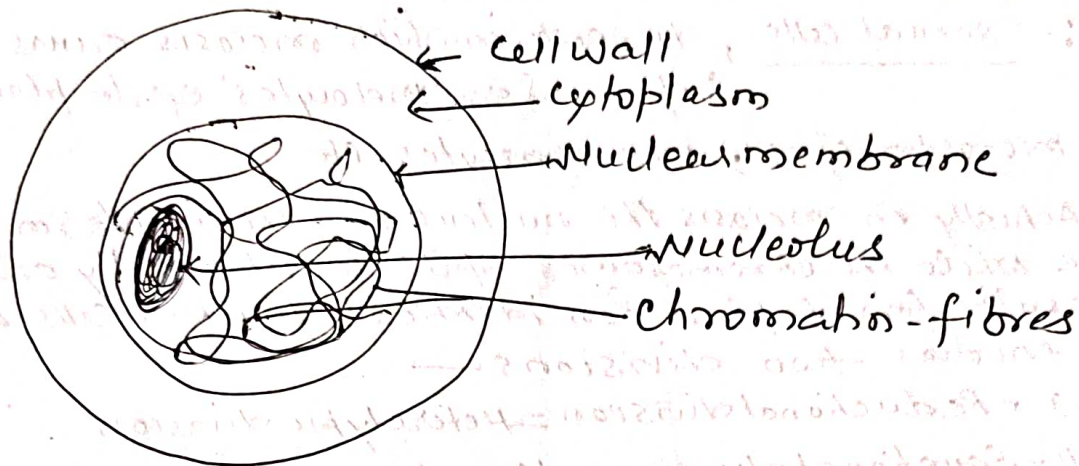
Cell-cycle showing an outline of Interphase and meiosis

INTERPHASE :- It can be studied under —

- (i) G₁-phase — RNA + Protein are synthesized
- (ii) S-Phase = Synthesis-Phase — Here, replication of DNA or doubling of chromosomes takes place.

(2)

Here, "S-phase" is slightly longer than that of mitotic "S-phase" and once the very cell enters the S-phase, then that very cell is bound to divide as here duplication of chromosomes takes place.



INTERPHASE

MEIOSIS-I-

It can be studied under_____

(A) PROPHASE-I ∴ It is so complicated hence can be studied under _____

(1) Leptonema / Leptotene

Gr. words — $\left\{ \begin{array}{l} \rightarrow \text{Leptos} = \text{thin} \\ \rightarrow \text{nema} = \text{thread} \end{array} \right.$

It is the 1st stage of Prophase-I and is characterized by _____

(2) The nuclear membrane and the nucleolus remain intact.

(ii) Chromomeres start to appear along the ~~thin~~ length of the thread like chromosomes.

(iii) The number and the position of Chromomeres remain constant for homologous chromosomes.

(2) Zygonema / Zygotene

Gr. words — $\rightarrow$ 2ygon = adjoining
 $\rightarrow$ nema = thread

It is also known as pairip stage and is characterized by _____

(ii) The homologous chromosomes become so closely approximated to each other due to mutual attraction between similar allelic genes and

- (3)
- process is being referred as "synapsis".
- (ii) The synapsed or paired chromosomes are known as bivalent.
- [Actually in synapsed chromosomes i.e. bivalent, the homologous chromosomes keep apart by 0.2 micron which is filled up with proteinaceous substance known as synaptonemal complex.]
- (iv) The nuclear membrane and the nucleolus remain intact (well-visible).

(3) Pachynema / Pachytene

Gr. words — $\rightarrow$ Pachus = Thick
 $\rightarrow$ nema = thread.

It is the 3rd stage of Prophase-I and is characterized by _____

- (i) The more condensed (thick) and short bivalents appear.
- (ii) Each chromosome of bivalent splits longitudinally to form tetrad stage of chromatids.
- (iii) Crossing over takes place. [Crossing-over is a phenomenon during which exchange of parts occurs between non-sister chromatids of the homologous chromosomes. Actually during longitudinal splitting one or more breakage may occur ~~at~~ at the same level in adjacent non-sister chromatids of homologous chromosomes by "Endonuclease" as reported by Stern and Hotta, 1968, these breakage immediately followed by exchange of parts ~~that~~ seals up by "Ligase" and the process is being referred as crossing-over].

(4) Diplonema / Diplotene

Gr. words — $\rightarrow$ Dia = double
 $\rightarrow$ nema = thread.

It is characterized by _____

- (i) The immediately synapsed or paired chromosomes, now start to separate but fail to separate at the point of exchange (crossing over) resulting into either cross (X) or ring or chain like configuration known as Chiasma/chiasmata.
- (ii) Chiasma/chiasmata are consequent result of crossing-over.

- (iii) The nuclear membrane and the nucleolus remain intact. (iv) At late diplotene terminalization begins.

[Terminalization :- The movement of chiasma/chiasmata along the length of chromosomes and finally slip off is known as terminalization].

(5) Diakinesis

It is characterized by —

- (i) Terminalization is completed.
- (ii) The nuclear membrane and the nucleolus are completely disorganized.
- (iii) Initiation of spindle fibres begins from respective poles by the activities of microtubules.

METAPHASE-I

It is characterized by —

- (i) The nuclear membrane and the nucleolus remain disorganized.
- (ii) The chromosomes arranged themselves at the equator of the cell more or less in two rows i.e. bivalents are arranged at the equator in single row.
- (iii) The spindle fibres get attached with the centromere of respective chromosomes of bivalents.

ANAPHASE-I

It is characterized by —

- (i) The nuclear membrane and the nucleolus remain disorganized.
- (ii) Due to contraction of spindle fibres the entire chromosomes i.e. along with its both chromatids move to their respective poles, (at the rate 1 micron/minute and total displacement i.e. from equator to respective poles require 30 ATP molecules), giving generally V or U-shaped appearance but $\checkmark$ or L-shaped is not uncommon.

TELOPHASE-I

(5)

Gr-words → Telos = terminal/last
→ Phase = stage

characterized by —

- (i) The arrival of chromosomes to their respective poles marks the initiation of Telophase-I.
- (ii) The nuclear membrane reorganized but the nucleolus can't i.e. remain disorganized.
- (iii) Remains of spindle fibres are present.

INTERKINESIS:-

After Telophase-I, the very cell enters a short period of rest known as INTERKINESIS. It is oftenly regarded as INTERPHASE-II but it is quite different from that of INTERPHASE-I as here no doubling of chromosomes occurs, after that the very cell enters meiosis-II.

MEIOSIS-II

[It is just similar to that of mitosis and can be well-illustrated with diagrams]

PROPHASE-II :- Nuclear membrane disorganized and initiation of spindle fibres starts just reverse to the 1st one.

METAPHASE-II

Chromosomes lie at the equator of cell and spindle fibres get attached with centromere of each chromosomes from both poles.

ANAPHASE-II

Due to contraction of spindle fibres and centromeric division chromatids separate and move to their respective poles.

TELOPHASE-II

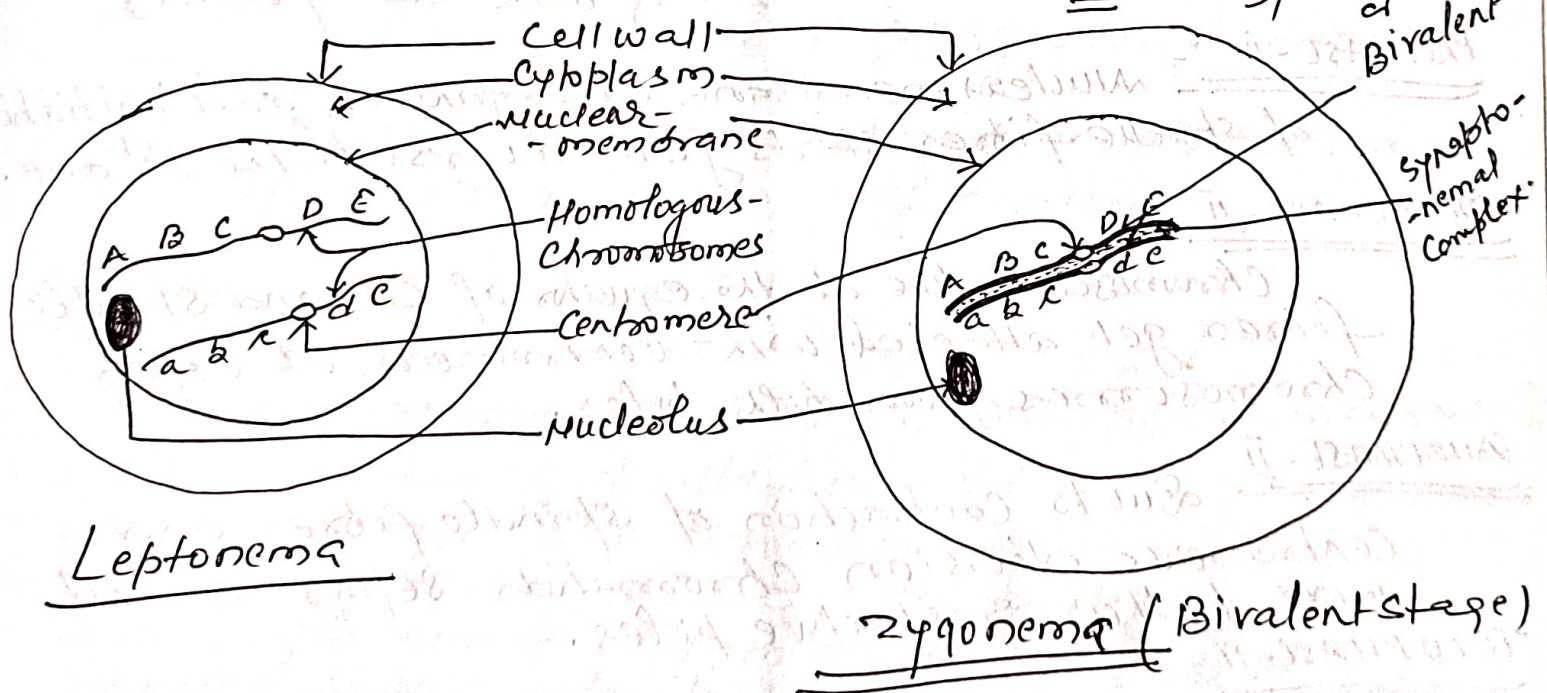
- Daughter chromosomes (chromatids) reach to their respective poles.
- The nuclear membrane and the nucleolus reorganized
- Spindle fibres are completely disorganized.

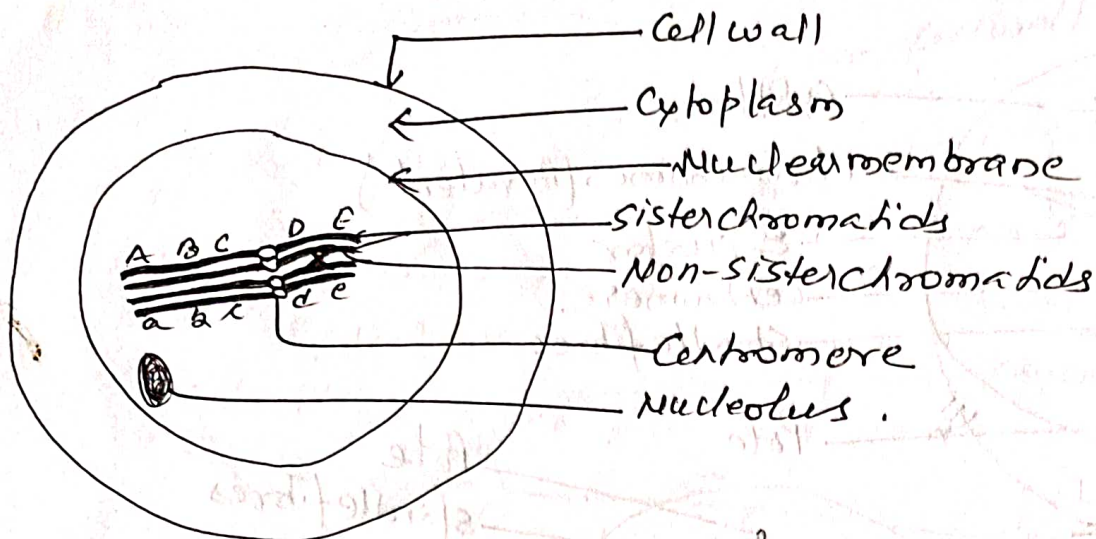
CYTOKINESIS :— i.e. division of cytoplasm. ⑥

In some cases cytokinesis ^{occurs} immediately ~~after~~ after Telophase-I by cell plate formation in plants whereas by membrane constriction in animal cells but in majority of cases especially in plants cytokinesis doesn't occur till the meiosis-II is completed. And as a result 4-haploid non-identical daughter cells are formed.

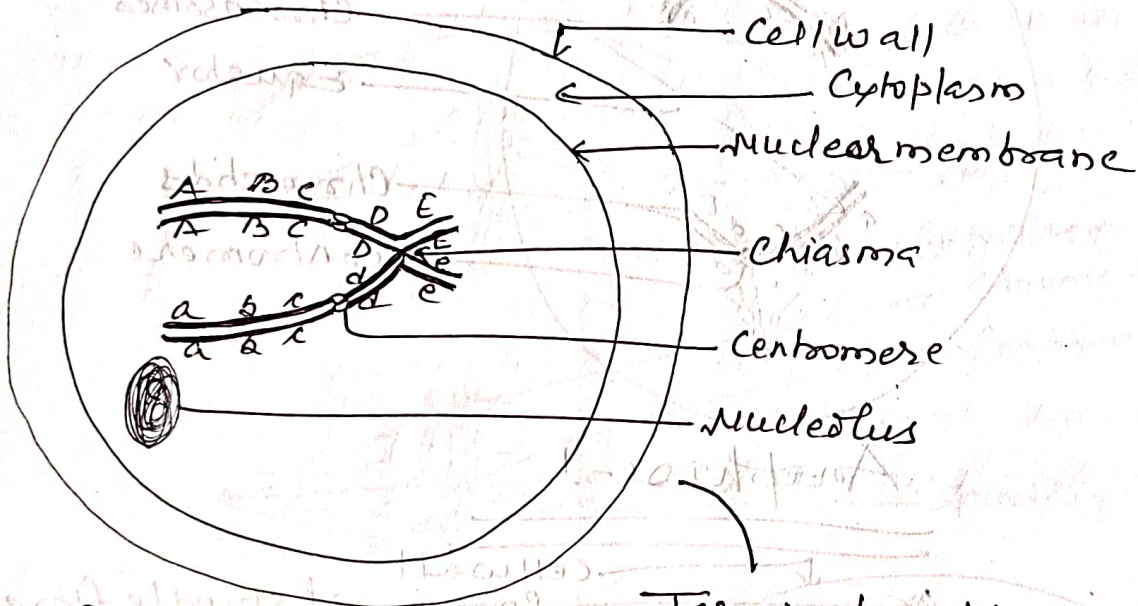
Significance of meiosis :-

- (i) Due to meiosis chromosome number just reduce to half in their daughter cells which further ensure us the chromosome number remain constant in sexually reproducing organisms even after fertilization has been taken place.
- (ii) As crossing over occurs in Pachytene of Prophase-I, which leads the formation of "recombinant chromosomes" (new combination of characters) responsible for variations in the offsprings (progeny) and these variations act as raw material for further evolution.



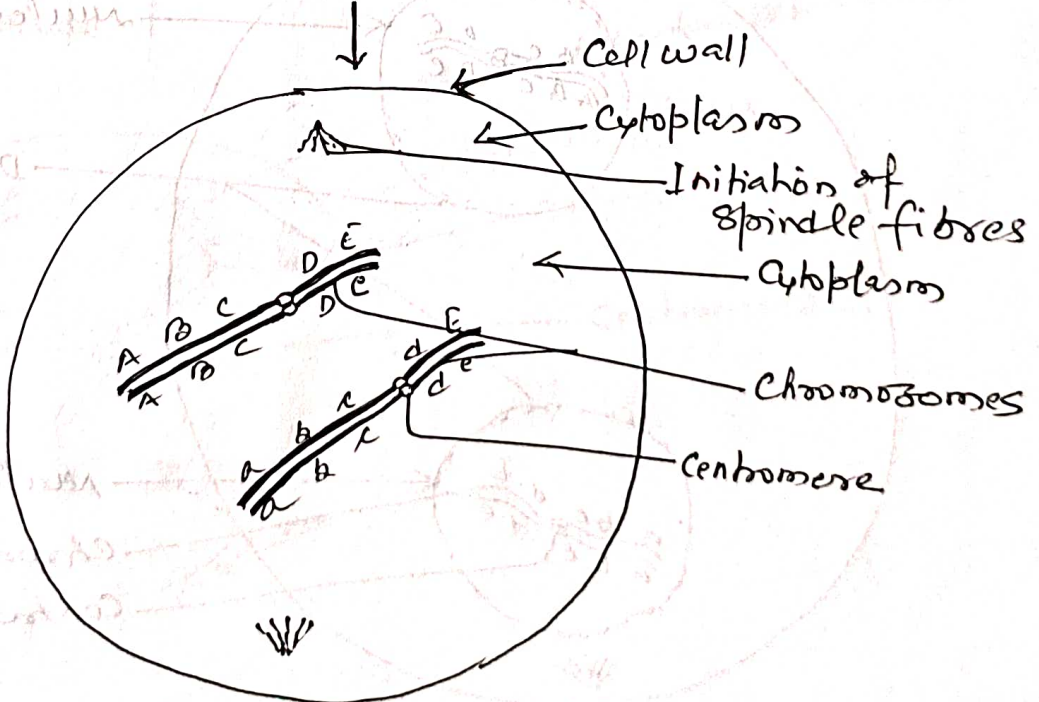


Pachynema (Tetrad stage of chromatids)



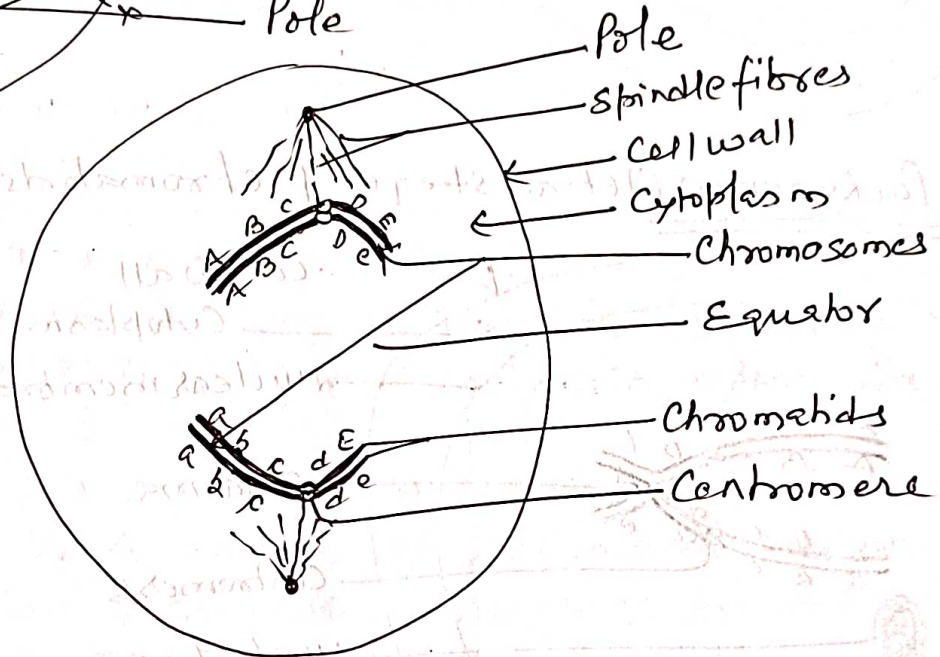
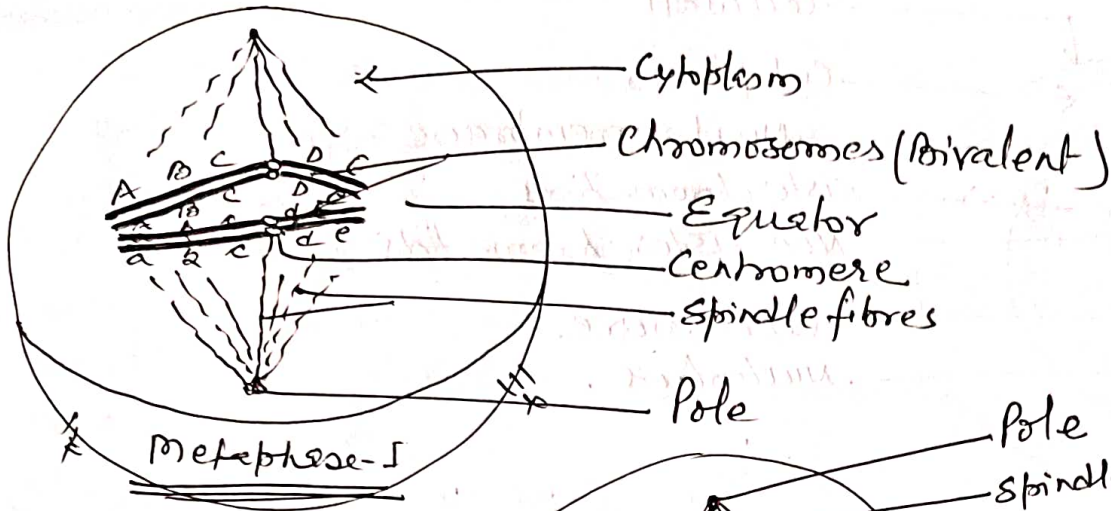
Diplonema

Terminalization

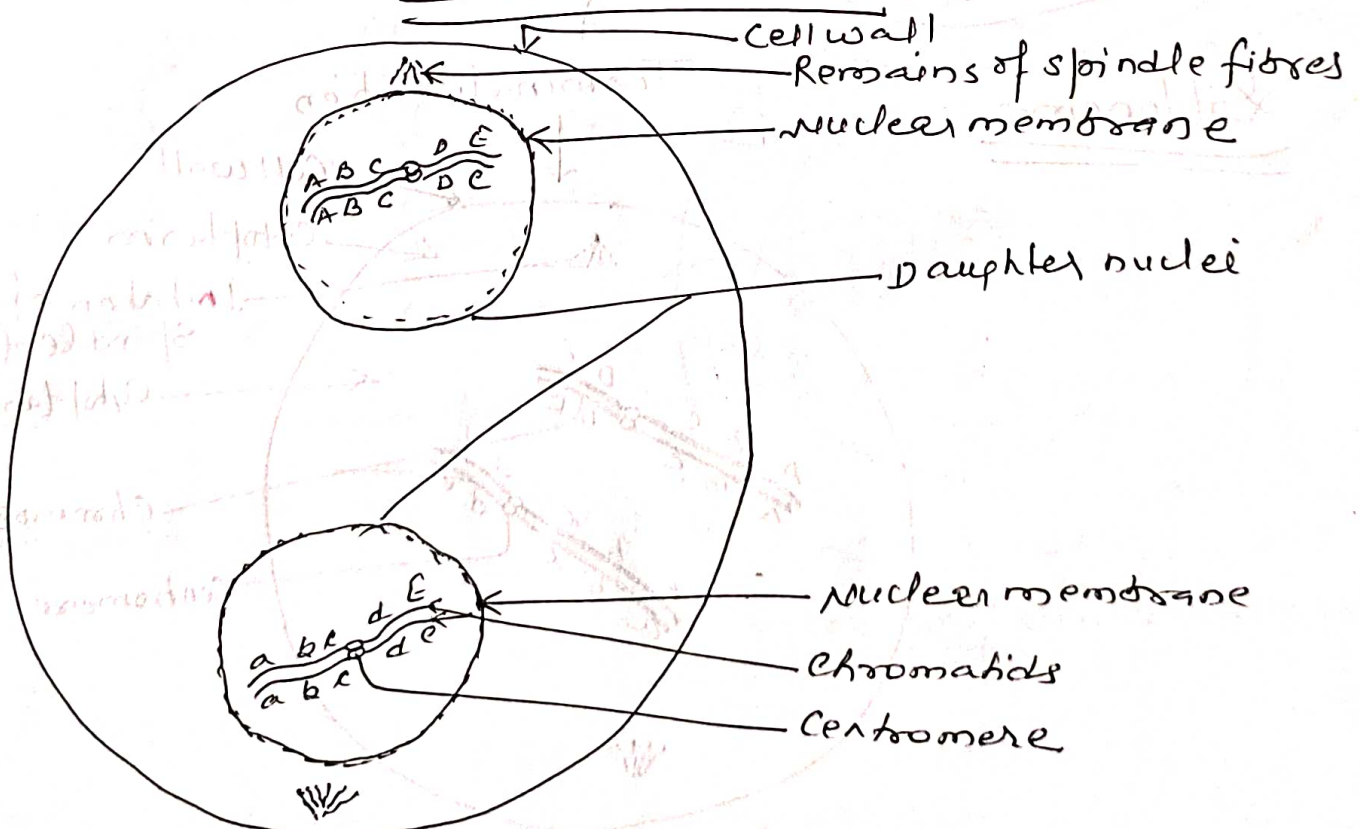


Diakinesis

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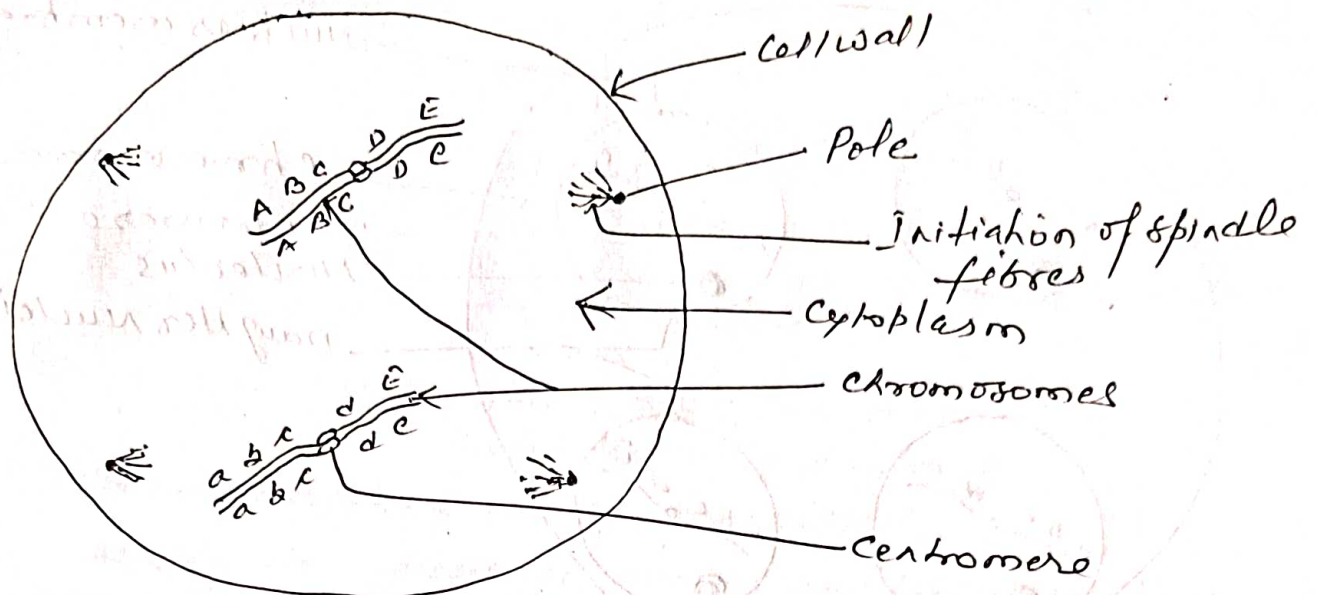


Anaphase-I

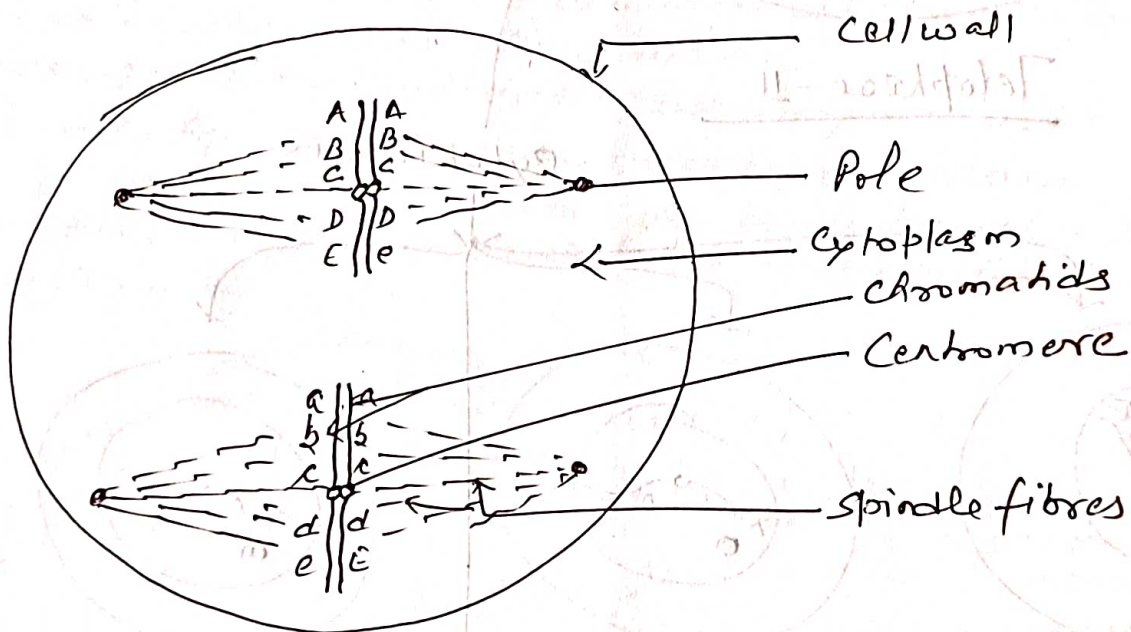


Telophase-I

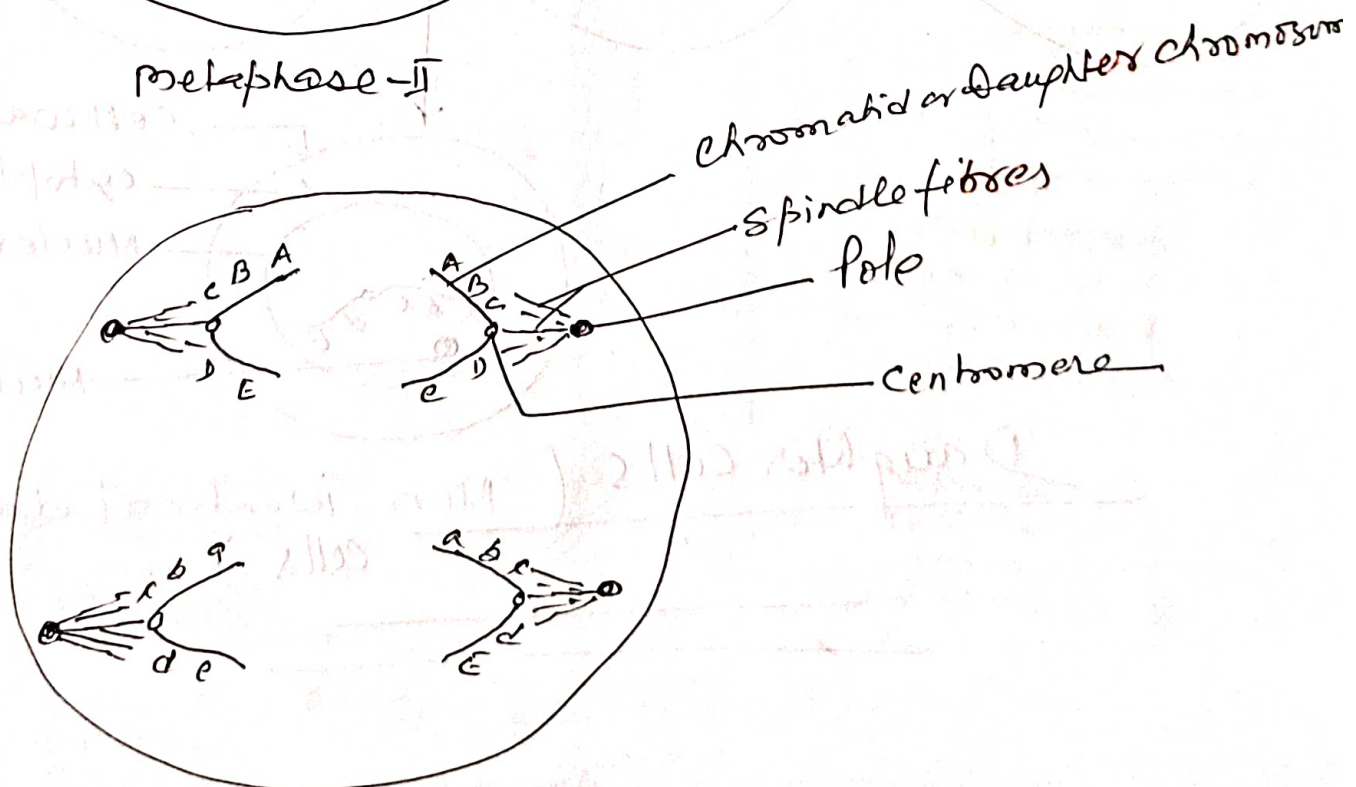
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Prophase-II

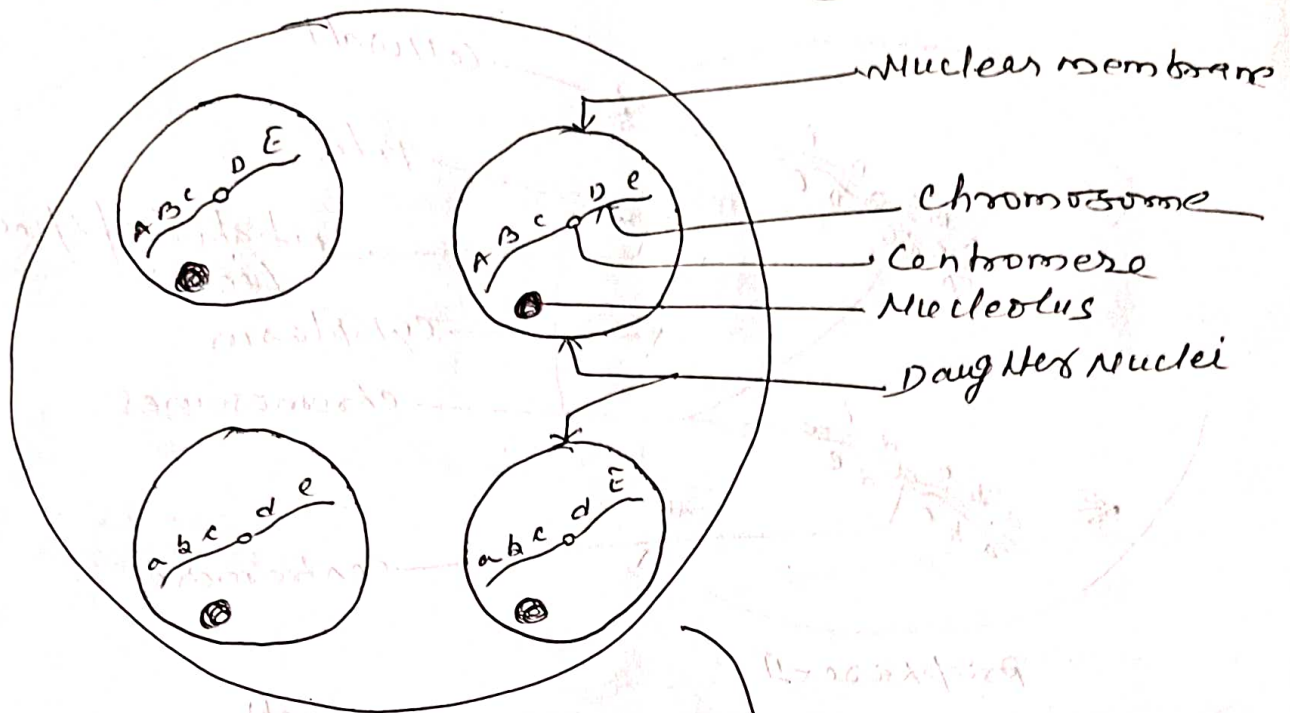


Metaphase-II



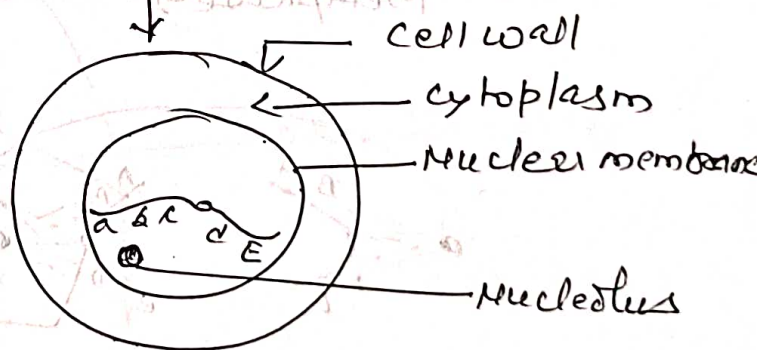
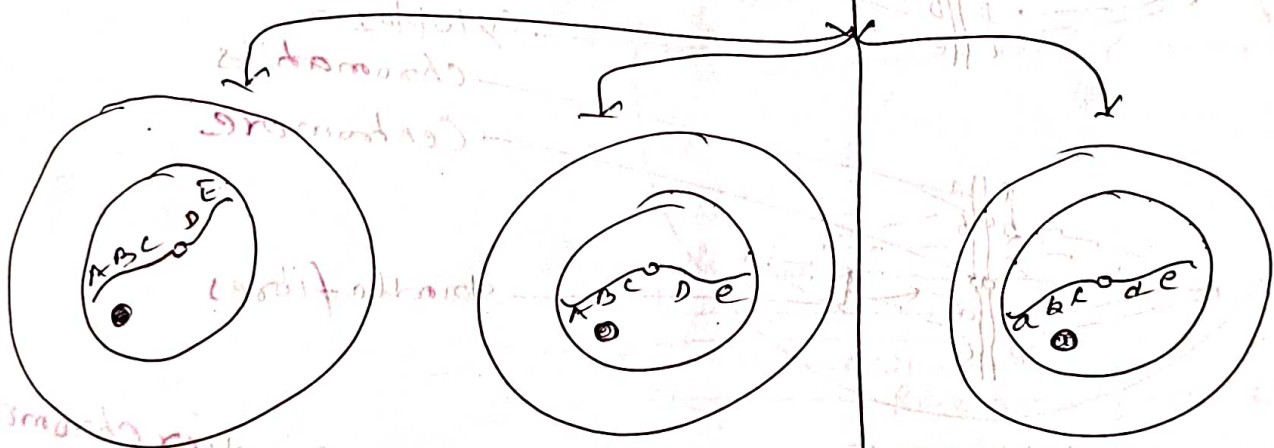
Anaphase-II

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Telophase-II

Cytokinesis



Daughter cells (Non-identical daughter cells)